

Shaan Bawa

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EDUCATION

University of Kansas

Lawrence, Kansas

Bachelor of Science in Computer Science | Exp. Graduation: May 2028 | GPA: 3.96

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Database Systems, Embedded Systems, Operating Systems (Current), Cyber Defense (Current), Theory of Computing (Current)
- **Honors:** Upsilon Pi Epsilon, University Honors Program, Dean's List, Kansas Governor's Scholar, Hixson Scholar.

WORK EXPERIENCE

Rock City Motel | Minneapolis, Kansas

Software Engineer Intern | May 2026 - August 2026

- Architected and deployed a dual-database Property Management System (React, TypeScript, Next.js, Fastify, PostgreSQL) to manage a 16-room commercial property, closing 29 full-lifecycle GitHub issues.
- Shipped a public direct-booking site on Next.js/Vercel with a Stripe PCI-compliant checkout, processing \$10,000+ in revenue while bridging cloud and LAN databases via a custom import-on-assign architecture.
- Engineered a deterministic availability engine utilizing PostgreSQL EXCLUDE constraints, resulting in 0 double-bookings and 100% accurate room assignments since deployment, backed by 44 Vitest tests.
- Diagnosed and resolved a live production data-integrity bug where concurrent walk-in guests mutated a shared template row; implemented a secure fork-on-edit database pattern with regression tests.
- Accelerated full-stack development, database migration, and complex debugging by integrating Agentic AI tooling (Cursor/Antigravity) as a default part of the daily engineering workflow.

Front Desk & Operations Associate | October 2019 - August 2024

- Managed daily front desk operations and payment processing, resolving guest inquiries to ensure high customer satisfaction during peak seasons.
- Served as the on-site technical lead, troubleshooting network hardware issues and maintaining booking software connectivity with third-party vendors (Expedia, Booking.com).

PROJECTS

Uni-League – Personal Project, Dec 2025 – June 2026

- Engineered a high-performance web application using JavaScript (React) and Python (Flask), ensuring rigorous attention to semantic HTML structure and component hierarchy.
- Implemented functional programming patterns to manage state and complex data flows from the Riot Games API.
- Developed custom algorithms to calculate and sort player leaderboards efficiently.
- Designed a relational data schema to track player statistics, ensuring data consistency and validity across the full-stack application.

BetOnMe, HackKU26 – HackKU26, April 2026 | Winner: MLH Track

- Engineered the backend API architecture using Next.js to power a decentralized accountability platform, winning an MLH-sponsored track for secure ledger integrations.
- Built an automated proof-verification pipeline by integrating the Gemini Vision API with custom EXIF/GPS metadata parsers to validate user location and timestamp data.
- Implemented complex on-chain escrow logic via the XRP Ledger, synchronizing transactional state with a MongoDB database to securely and autonomously route failed stakes to charity.

The Sopranos Syndicate Tracker – March 2026 – May 2026

- Architected a web application using ASP.NET Core MVC to simulate an FBI wiretap dashboard, managing complex data hierarchies for organized crime entities.
- Developed Domain Models and Controllers in C# to handle business logic and interface with a MySQL database.
- Designed a relational schema with 6 interconnected tables, utilizing SQL inner joins and aggregations to generate hierarchical financial reports.

ACTIVITIES

Kansas Super Computing Club | Treasurer & Cluster Competition Member | Aug 2024 - Present

- Built and benchmarked a Linux-based computing cluster for regional high-performance computing competitions, optimizing workflow processes using command-line tools.

SKILLS & INTERESTS

Technical Skills: JavaScript/TypeScript (React, Next.js, Vite), Python (Flask), C# (ASP.NET Core), SQL (PostgreSQL, MySQL, Drizzle ORM).

AI & Architecture: Agentic AI Workflows (Cursor), LLM Integrations (Gemini API), REST APIs, JSON, Docker, Git.